

# Synthetic Market Phase 21

Source-driven restoration across two axes on the compiled patch path: Leader (L4, top-4 components) and Dispersion (L35, top-4 components). This report replaces the earlier mixed-batch draft and uses only the validated `batch_size = 16` reruns: 48 matched pairs per axis, separately generated source behaviors, and a `swap_components` intervention that inserts source-side coefficients into the base prompt.

| DATE         | ROWS        | PATCH                                      | ENGINE                       |
|--------------|-------------|--------------------------------------------|------------------------------|
| 3 April 2026 | 48 per axis | swap_components, source-driven donor means | compiled non-eager custom-op |

MAIN READ

The validated bs16 rerun preserves the main scientific split from the old draft while fixing the decoding pathology. Leader still shows the stronger partial restoration signal on action choice and spend. Dispersion is no longer obviously broken at the decode level, but it remains weak on action restoration and poor on response length. Phase 22 should therefore stay leader-first rather than promoting dispersion into the main path-validation target.

| LEADER<br>RESTORE       | TOOL-TOKEN | DISPERSION<br>TOOL-TO-<br>KEN RESTORE | LEADER<br>RESTORATION                                                 | LENGTH | DISPERSION<br>RESTORATION      | LENGTH |
|-------------------------|------------|---------------------------------------|-----------------------------------------------------------------------|--------|--------------------------------|--------|
| 0.25                    |            | 0.136                                 | -1.396                                                                |        | -3.13                          |        |
| 4 of 16 restorable rows |            | 3 of 22 restorable rows               | negative means the patched response moved away from the source length |        | much more negative than Leader |        |

## What Was Tested

For each matched pair, the run produced three behaviors:

- the source prompt on its own
- the base prompt on its own
- the base prompt with a `swap_components` patch

The question is simple:

- did the patch make the base prompt behave more like its matched source prompt?

This report uses denoise pairs. In plain English, that means:

- the base row is the lower-valued member of the pair on the chosen metric
- the source row is the higher-valued member of the pair

Here the chosen metric depends on the axis:

- Leader uses `vol_1h_max`
- Dispersion uses `pct_1h_mad`

`swap_components` also has a plain-English meaning here:

- it does *not* copy every hidden-state token from source to base
- it captures the source row, averages over the market span, and extracts the selected coefficients
- it then writes those source-side coefficients into the base row over the base market span

One operational note matters for interpretation:

- both axes were rerun at `batch_size = 16`
- both used the same compiled non-eager custom-op patch path
- this replaces the earlier draft whose Dispersion result was contaminated by the batch-32 tool-surface failure

## Decode Validity

Before interpreting the restoration metrics, the reruns need to clear the decode-validity check that the old draft failed.

| Check                                    | Leader                  | Dispersion              |
|------------------------------------------|-------------------------|-------------------------|
| Base rows with valid tool calls          | 48 / 48                 | 48 / 48                 |
| Source rows with valid tool calls        | 48 / 48                 | 48 / 48                 |
| Finish reason stop                       | 48 / 48 base and source | 48 / 48 base and source |
| 15000-token cap hits                     | 0                       | 0                       |
| !!!! corruption rows                     | 0                       | 0                       |
| Missing <code>&lt;/think&gt;</code> rows | 0                       | 0                       |

So the key methodological upgrade over the old report is not only the science. It is that both axes are now being evaluated on a clean, validated generation regime.

## Axis Comparison

| Measure                | Leader                | Dispersion        | How to read it                    |
|------------------------|-----------------------|-------------------|-----------------------------------|
| Source tool-name match | 0.979 $\rightarrow$ 1 | 1 $\rightarrow$ 1 | Leader improves from near-satura- |

| Measure                            | Leader        | Dispersion    | How to read it                                                                                         |
|------------------------------------|---------------|---------------|--------------------------------------------------------------------------------------------------------|
|                                    |               |               | tion; Dispersion stays saturated                                                                       |
| Source tool-token match            | 0.667 → 0.708 | 0.542 → 0.521 | Leader improves; Dispersion is slightly negative even after the clean rerun                            |
| Tool-token restoration rate        | 0.25          | 0.136         | Higher is better on rows that needed fixing                                                            |
| Tool-token backfire rate           | 0.063         | 0.154         | Lower is better on rows that were already correct                                                      |
| Spend improvement rate             | 0.667         | 0.6           | Both axes move spend toward the source more often than not; Leader still has the clearer action result |
| Generated-token improvement rate   | 0.438         | 0.396         | Both are weak; Leader is still less bad                                                                |
| Mean normalized length restoration | -1.396        | -3.13         | Negative means the patch moved away from the source                                                    |

## Restorable Rows And Backfires

| Metric           | Leader                                      | Dispersion                                  |
|------------------|---------------------------------------------|---------------------------------------------|
| Tool name        | restorable 1 restored 1 backfires 0 / 47    | restorable 0 restored 0 backfires 0 / 48    |
| Tool token       | restorable 16 restored 4 backfires 2 / 32   | restorable 22 restored 3 backfires 4 / 26   |
| Spend pct        | restorable 3 improved 2 backfires 2 / 47    | restorable 5 improved 3 backfires 1 / 48    |
| Generated tokens | restorable 48 improved 21 backfires 27 / 48 | restorable 48 improved 19 backfires 29 / 48 |

This table is the clearest practical summary:

- Leader helps on tool-token choice more often than it hurts
- Dispersion hurts tool-token choice almost as often as it helps

- Dispersion spend is now real on a small five-row restorable subset, but the action-surface result is still weak
- neither axis gives a clean length-restoration story

## Mechanical Validity

| Measure               | Leader | Dispersion |
|-----------------------|--------|------------|
| Rows with patch stats | 48     | 48         |
| Patch applied rate    | 1      | 1          |
| Patch skipped rate    | 0      | 0          |

Mechanically, both runs are clean. Every row has patch diagnostics and nothing was skipped.

## Methodology Scorecard

| Best-practice item                                               | Status  |
|------------------------------------------------------------------|---------|
| Matched prompt pairs and separately generated source behaviors   | meets   |
| Source-driven coefficient swap rather than generic ablation      | meets   |
| Compiled non-eager custom-op execution path for the patch itself | meets   |
| Per-row patch diagnostics with zero skips on both axes           | meets   |
| Exact token-by-token source activation transplant                | missing |
| Clean restoration of action choice on every tested axis          | partial |
| Clean restoration of response length / verbosity                 | missing |
| Path-level validation                                            | missing |

The main methodological upgrade over Phase 20 is that Phase 21 uses *generated source behaviors* and a *source-driven swap*. That makes this a real restoration test, not only a disruption test.

The main methodological upgrade over the earlier Phase 21 draft is that this version also clears a *decode-validity* check: both axes were rerun in a batch regime with no degenerate decodes, so the axis comparison is now operationally apples-to-apples.

## Interpretation

Leader and Dispersion still do not behave the same way.

Leader is the better causal candidate:

- it improves source match on the action surface we care about most
- tool-token restoration is materially above backfire
- spend moves toward the source more often than not

Dispersion is cleaner than it looked in the old draft, but it is still weaker:

- tool-name match is saturated, so there is no tool-name restoration story to tell

- tool-token match is slightly worse overall after the patch
- tool-token restoration and backfire are roughly the same size
- spend does move toward the source on a five-row restorable subset, but that is not enough to promote Dispersion into the main causal handle

The combined takeaway is not “restoration failed.” It is more specific:

- the market representation is not uniform
- Leader looks more like a usable causal handle on action choice
- Dispersion may still matter for a narrower calibration-like quantity, but not for clean action restoration
- neither axis yet restores whole-response length cleanly

## Next Step

The next experiment should push the restoration story where it is currently strongest:

- keep `batch_size = 16` for this Qwen3 thinking + forced-tool setup unless batch invariance proves 32 stable
- run Phase 22 path validation on *Leader first*
- keep Dispersion as a contrast case or calibration follow-up, not as the primary Phase 22 path-validation target
- if possible, move from span-mean coefficient swapping toward a more exact source-activation transplant

If Leader keeps restoring tool-token choice under those stronger tests, the causal claim becomes materially harder to dismiss.